```
#include <iostream>
#include <cmath>
using namespace std;
double f(double x) {
  return x * x * x - 10 * x * x - 5 * x + 5;
}
double regulaFalsi(double a, double b, double tolerance, int max_iterations) {
  if (f(a) * f(b) >= 0) {
    cout << "The method requires that f(a) and f(b) have different signs." << endl;
    return NAN;
  }
  double c;
  for (int i = 0; i < max_iterations; i++) {
    c = b - (f(b) * (a - b)) / (f(a) - f(b));
        if (fabs(f(c)) < tolerance) {
      break;
    }
    if (f(c) * f(a) < 0) {
      b = c;
    } else {
      a = c;
    }
  }
  return c;
}
int main() {
  double a, b, tolerance;
  int max_iterations;
  cout << "Enter the interval [a, b]: ";
  cin >> a >> b;
  cout << "Enter tolerance: ";
  cin >> tolerance;
  cout << "Enter maximum iterations: ";
  cin >> max_iterations;
  double root = regulaFalsi(a, b, tolerance, max_iterations);
  if (!isnan(root)) {
    cout << "The root is approximately: " << root << endl;
  }
  return 0;
}
```